GGG-S-595B
February 9, 1972
SUPERSEDING
Fed. Spec. GGG-S-595A
June 28, 1961

FEDERAL SPECIFICATION

SOLDERING IRONS, NONELECTRIC, (COPPERS)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers nonelectric soldering irons, commonly referred to as "coppers".

1.2 Classification.

1.2.1 Types and classes. Nonelectric soldering irons (coppers) shall be of the following types and classes, as specified (see 6.2):

Type I. - Solid copper.

Class 1. - Regular tinners' point (pyramid).
Class 2. - Flat point (wedge).

Type II. - Composite copper and steel.

Class 1. - Regular tinners' point (pyramid).
Class 2. - Flat point (wedge).

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 3439

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Federal Standard:

Fed. Std. No. 123	- Marking for Domestic Shipment (Civil Agencies).

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(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-W-45562	- Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

(Copies of Military Specifications, Standards, Drawings, and Publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Illustrations. The illustrations shown herein are for convenience of identification and are not intended to preclude the purchase of soldering irons which otherwise conform to the requirements of this specification.

3.2 Materials. Materials shall be as specified herein. Materials not specifically designated shall be of good commercial quality. Copper shall be commercially pure, cast or wrought, and free of cracks, blow-holes, seams, and other imperfections.

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3.3 Size. Soldering irons shall be furnished in the sizes specified (see 6.2). Size is defined as the weight per single soldering iron (copper) including the handle rod, and shall be in accordance with table I for type I and table II for type II.

3.4 Wooden handle. Unless otherwise specified (see 6.2), the soldering irons, except for the 3-ounce type I iron, shall be furnished without handles. When handles are specified they shall be made of straight grained, sound, well seasoned hardwood. They shall be of conventional shape and design, proportioned to afford a comfortable handgrip.

3.5 Type I. Type I soldering irons shall consist essentially of a copper body and a steel handle rod. They shall conform to table I and be similar to figure 1.

TABLE I. Type I, class 1, regular tinners' point (pyramid) and class 2, flat point (wedge)

Size (wt. of single copper)		Length of body		Thickness of body dimension "C"	Handle rod	
		Overall dimension "A"	Pointed dimension "B"		Diameter dimension "D"	Length dimension "E"
+15 -0 percent	+15 -0 percent	±5 percent	±10 percent	±10 percent	+25 -5 percent	±10 percent
Ounces	Pounds	Inches	Inches	Inches	Inches	Inches
3	--	2 3/8	1	1/2	3/16	(see Note 1)
--	1/2	3 1/8	1	11/16	9/32	8 1/4
--	1	3 3/4	1 1/8	29/32	7/16	8 1/4
--	1 1/2	4 1/2	1 7/16	1 1/32	7/16	8 1/4
--	2	4 15/16	1 25/32	1 1/4	17/32	9
--	2 1/2	5 3/16	1 25/32	1 1/4	9/16	9
--	3	5 7/8	1 31/32	1 3/8	9/16	10
--	4	6 1/2	2 5/32	1 1/2	9/16	11
--	5	6 1/2	2 11/32	1 5/8	9/16	11 1/4
--	6	7	2 1/2	1 3/4	9/16	11 5/8
--	7	7 3/8	2 11/16	1 7/8	11/16	11 5/8

NOTE 1. The 3-ounce size shall have an overall length of 11 1/2 inches (including handle).

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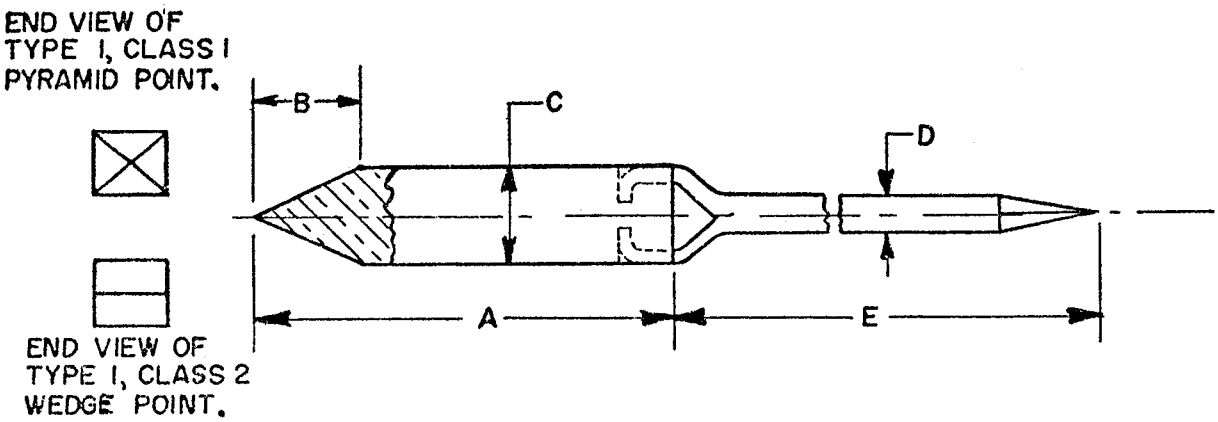


FIGURE 1. - Type I, class 1, regular tinners' point (pyramid) and class 2, flat point (wedge).

3.5.1 Body. The body shall be of solid copper approximately square in cross section, with chamfered edges.

3.5.1.1 Class 1. The point of class 1 irons shall be in the shape of a square pyramid with the apex blunted.

3.5.1.2 Class 2. The point of class 2 irons shall have two sides uniformly sloped to form a wedge shaped point with the sharp edge removed. The width of the point shall be equal to the width of the copper.

3.5.2 Handle rod. The handle rod shall have one end pointed for insertion into a wooden handle. The other end of the handle rod, except for the 3-ounce size, shall be securely attached to the copper by passing the forked steel into the copper at right angles to the longitudinal axis and then closing the fork by welding or brazing. The fork shall be so attached that its sides will be imbedded in the copper and will not project beyond the lateral surfaces. The handle rod for the 3-ounce size shall have one end securely fixed in the end of the copper body and the other end fixed in a wooden handle (see 3.4).

3.6 Type II. Type II soldering irons shall consist essentially of a composite copper and steel body and a steel handle rod. They shall conform to table II and be similar to figure 2.

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TABLE II. Type II, class 1, regular tinners' point (pyramid) and class 2, flat point (wedge)

Size (wt. of single copper)	Length of body			Body diameter dimension "D"	Thread depth dimension "E"	Thread diameter[2] dimension "F"	Handle rod	
	Copper portion dimension "A"	Steel portion dimension "B"	Pointed portion dimension "C"				Diameter dimension "G"	Length dimension "H"
	+15 -0 percent	±10 percent	±10 percent	±10 percent	+20 -0 percent	±10 percent	+25 -5 percent	±10 percent
Pounds	Inches	Inches	Inches	Inches	Inches	Inches	Inches	Inches
1/4[1]	1 1/4	3/4	3/4	5/8	5/16	5/16	1/4	7
3/4[1]	2	1	1 1/4	1	7/16	7/16	3/8	8
1	2 3/8	1 3/8	1 5/8	1 1/8	1/2	1/2	7/16	8 1/4
2	2 21/32	2 19/32	2	1 3/8	5/8	5/8	17/32	9
5	3 11/32	2 29/32	2 11/16	1 7/8	3/4	3/4	9/16	11 1/4

[1] Class 1 only.

[2] All threaded parts shall be of a nationally recognized commercial standard thread having standard size, shape, and tolerances in order to insure correct mating of parts and convenience of interchangeability and replacement.

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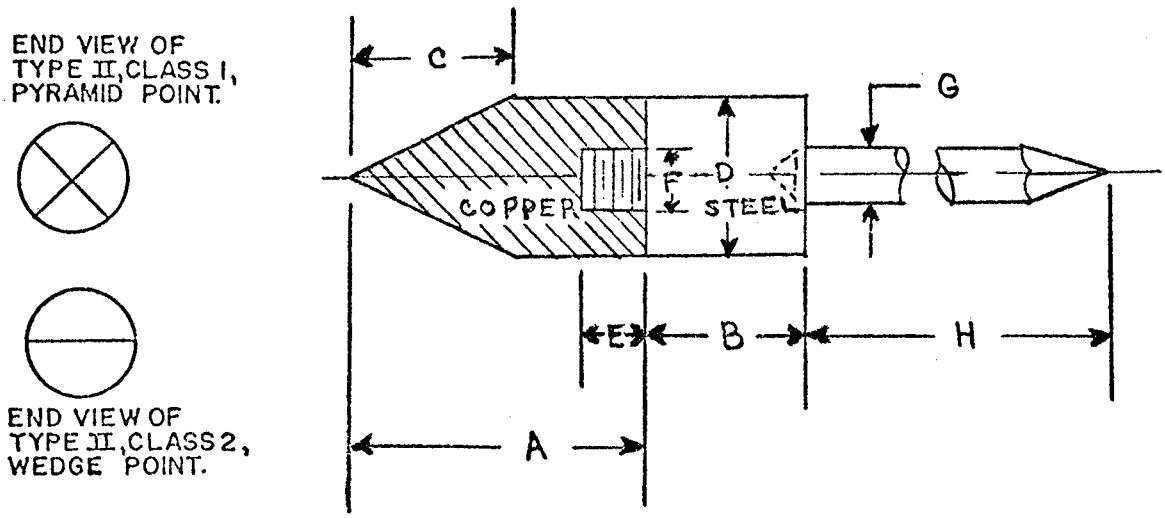


FIGURE 2. - Type II, class 1, regular tinners' point (pyramid) and class 2, flat point (wedge).

3.6.1 Body. The body shall be made in two sections; a tip section made of copper and a stud section made of steel. These two sections shall be joined together by means of an integral threaded steel stud. The cross section of the body for all sizes except the 1/4-pound and the 3/4-pound sizes of class 1 shall be round. The cross section of the body for the 1/4-pound and the 3/4-pound sizes of class 1 may be either round or hexagonal.

3.6.1.1 Class 1. The point of class 1 soldering irons shall be in the shape of a square pyramid, except that the 1/4-pound and 3/4-pound sizes shall have the apex blunted.

3.6.1.2 Class 2. The point of class 2 soldering irons shall have two sides uniformly sloped to form a wedge shaped point with the sharp edge removed. The width of the point shall be equal to the width of the copper.

3.6.2 Handle rod. The handle rod shall have one end pointed for insertion into a wooden handle. It shall be either an integral part of the steel section or a separate steel rod with one end securely fixed in a hole in the end of the steel section.

3.7 Identification marking. Each soldering iron (copper) shall be plainly and permanently marked with its size (weight per single iron) and with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

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3.8 Workmanship. Soldering irons shall be free of rust, scale, burrs, fins, and other defects which may adversely affect durability and serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Quality conformance inspection shall be conducted to the extent necessary for controlling product quality to assure that items offered for acceptance conform to all contractual requirements. As a minimum, quality conformance inspection shall consist of (a) and (b) as follows:

(a) Examination (see 4.3).

(b) Preparation for delivery examination (see 4.4).

4.3 Examination. Production items shall be examined to determine compliance with requirements of section 3. As a minimum, the product characteristics contained in 4.3.1 shall be examined.

4.3.1 Product characteristics.

Product characteristic	Requirement reference
Material not as specified.	(See 3.2).
Size not as specified.	(See 3.3).
Type and class not as specified.	(See 3.5 or 3.6).
Handle rod not as specified.	(See 3.5.2 or 3.6.2).
Identification marking not as specified.	(See 3.7).

4.4 Preparation for delivery examination. Preservation, packaging, packing, and marking for shipment and storage shall be examined to determine conformance with the requirements of section 5.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing, and marking. Unless otherwise specified (see 6.2), preservation, packaging, packing, and marking shall be in accordance with MIL-W-45562. The levels of preservation, packaging, and packing shall be as specified (see 6.2).

5.1.1 Civil agencies. In addition to any special marking specified in the contract or order, interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

6. NOTES

6.1 Intended use. The nonelectric soldering irons (coppers) covered by this specification are intended to be used for applying heat to melt solder and to heat metals that are to be joined by soldering.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number, and date of this specification.

(b) Type and class required (see 1.2.1).

(c) Size (weight of single iron) required (see 3.3 and tables I and II).

(d) Handles, if required (see 3.4).

(e) Preservation, packaging, packing, and marking, if different (see 5.1).

(f) Selection of applicable level of packaging and packing (see 5.1).

MILITARY CUSTODIANS:

Army - WC

Review activities:

Army - SM

User activities:

Army - None

Preparing activity:

Army - WC

CIVIL AGENCY INTEREST:

GSA-FSS

Project No. 3439-0204

Orders for this publication are to be placed with the General Services Administration, acting as an agent for the Superintendent of Documents. See section 2 of this specification to obtain extra copies and other documents referenced herein. Price 10 cents each.

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